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Unsupervised Depth Estimation Using Internal Calibration Parameters as Input

A camera-aware extension of Depth Anything V2 that reads the intrinsic matrix K directly — and an experimental campaign that proves, mechanistically and causally, that the model actually uses it.

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DEFENSE
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What this talk covers

I **Motivation & research questions** — why a single image cannot fix metric scale, and what the thesis asks.

II **Method** — the camera descriptor, the gated camera-token architecture, the training and loss design.

III **The experimental campaign** — two consequential failures found and fixed, a protocol pivot, and the causal chain of evidence from a correlational win to a mechanistic and causal proof.

the core of this talk

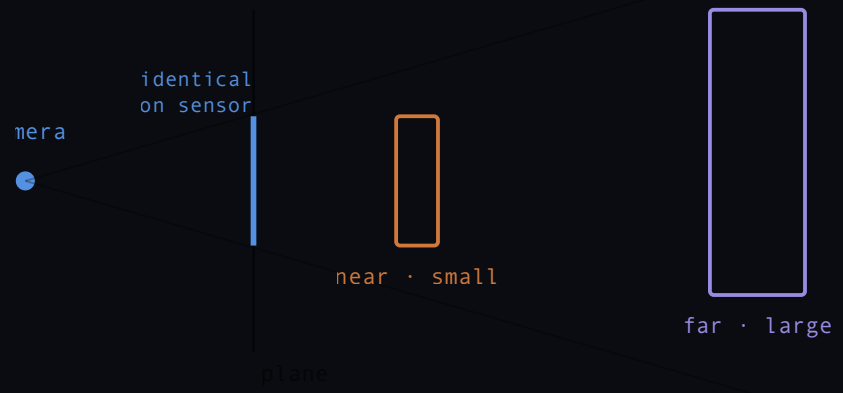
IV **Discussion & conclusions** — what was actually proven, what was not, and what comes next.

A single image does not determine scale

Two different worlds can cast the exact same image.

- Under pinhole projection, image position depends only on the **ratio** $X\!Z$ — a bigger object twice as far away is pixel-for-pixel indistinguishable from a smaller, closer one.
- Monocular depth models resolve this ambiguity **implicitly**, from learned priors about object size and perspective — priors that are wrong exactly when the scene doesn't match training statistics.
- The camera's **intrinsic matrix** K is metadata nearly every real camera already exposes; it supplies the missing focal-length/scale signal explicitly, while foundation depth models usually never see it.

Demo · little-big ambiguity. Open full-screen to see the complete 16:9 mobile video. Camera intrinsics supply an additional focal-length signal alongside the image.



$$u = f \cdot X / Z \quad — \quad \textit{same } u \textit{ for (small, near) and (large, far)}$$

Figure. Both rectangles subtend the same angle from the camera and land on the identical image-plane region — the image alone cannot tell them apart. The focal length f, part of K, supplies the geometric signal that relates image ratio to scene scale in this controlled ambiguity.

From hand-crafted cues to foundation models

Classical geometric cues

Hoiem et al. · Make3D (Saxena et al.) — convergent lines, texture, horizon. **Limit:** brittle scene-structure assumptions, poor generalization.

Supervised deep CNNs

Eigen et al. 2014 / 2015 — multi-scale, multi-task depth CNNs. **Limit:** learned scale tied to the training camera; evaluated mostly on normalized metrics.

Self-supervised stereo

Monodepth, Monodepth2 (Godard et al.) — view synthesis, $Z{=}fb/d$. **Limit:** scale comes from a fixed stereo baseline b, not from K.

Self-supervised video / SfM

Zhou et al. 2017 — K used only inside the warping function. **Limit:** the depth network itself never sees K; ambiguity remains.

Relative / foundation models

MiDaS, DPT, **Depth Anything V2** (this thesis's base) — excellent relative depth at scale. **Limit:** output is inherently non-metric.

Explicit-geometry outliers

Saxena et al. 2023 — FoV-conditioned diffusion (costly inference); MVSNet (Yao et al. 2018) uses camera calibration for multi-view cost volumes, not single-image depth.

Four ways to handle K — almost everyone picks the first three

1. **Ignored entirely** — classical geometric-cue and most supervised CNN methods.

2. **Implicitly absorbed into weights** — assumes K is constant across training and test; most relative/foundation models, including base Depth Anything V2.

3. **Used only inside the cost function** — self-supervised warping (Zhou et al.); the depth network itself never receives K as input.

4. **Explicit input conditioning** — the network's forward pass literally reads K. Rare in single-image depth (for example, CAM-Convs and UniDepth); MVSNet uses K but requires multiple calibrated views. **This thesis's position — for single-image, foundation-model depth.**

Three questions this thesis answers

RQ1 Do common monocular depth models behave **stably** as the dataset and the camera's intrinsic parameters change?

RQ2 Can **non-metric evaluation metrics conceal** instabilities that exist in the physical scale of predicted depth?

RQ3 Can incorporating **explicit camera-intrinsics** improve metric depth estimation and reduce a model's sensitivity to changes in the camera?

A hidden fourth requirement

Answering RQ3 with a benchmark win isn't enough — a win could be spurious. The thesis adds a **causal-validity requirement**: prove the model actually *reads* K, not merely correlates with it. That requirement is what the K-sweep and K-knockout tests in §III exist to satisfy.

II Method

Camera geometry · descriptor design · gated injection into Depth Anything V2 · training & losses

The pinhole model and K

A 3D point (X,Y,Z) projects to pixel (u,v) through the intrinsic matrix K — focal lengths f_x, f_y and principal point c_x, c_y. Every quantity a depth model must recover — absolute scale, object size in metres — is entangled with exactly these four numbers.

- known to the camera at capture time
- hidden from foundation depth models today

$$u = f_x \cdot X/Z + c_x$$

$$v = f_y \cdot Y/Z + c_y$$

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

A normalized, 9-dimensional camera descriptor

Raw (f_x, f_y, c_x, c_y) is resolution-dependent and hard for a network to consume directly — it is expanded into a normalized descriptor that also encodes field of view and image shape.

$$d(K,H,W) = \left[\tilde{f}_x, \tilde{f}_y, \tilde{c}_x, \tilde{c}_y, \quad \mathrm{FoV}_h/\pi, \mathrm{FoV}_w/\pi, \quad W/H, \quad \left(\frac{f_x + f_y}{2}\right)/\max(H,W), \quad H \cdot W / (1920 \cdot 1080) \right]$$

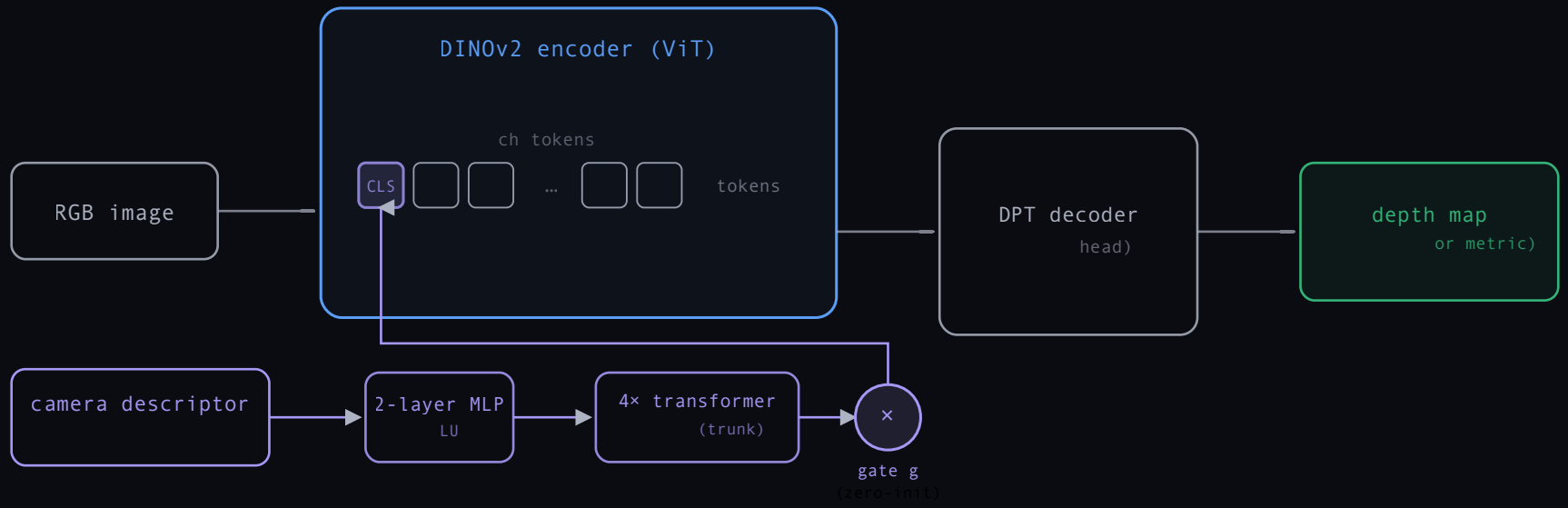
What each term contributes

Focal lengths & principal point (normalized) give raw calibration; field-of-view angles give a resolution-independent notion of "how wide is this shot"; aspect ratio and mean focal length help the encoder relate K to typical camera families it may have seen.

The coordinate-frame constraint

K and (H,W) must describe the **same** image frame — crop or resize the image without updating K to match, and the descriptor becomes geometrically wrong. This exact mistake is Bug #2 in the campaign, §III.

A gated camera token, injected into the DINOv2 encoder



Zero-initialized gate

g starts at exactly 0, so the camera token contributes nothing at step 0 — fine-tuning begins **identical** to the pretrained baseline and opens the gate only if K reduces the loss.

Frozen DINOv2 backbone

Only the DPT head and the camera encoder adapt (`lr 5e-6`, head 10× multiplier) — this preserves the ~62M-image pretrained prior and avoids the catastrophic zero-shot drift discussed in §III.

Why inject a token — not concatenate, not FiLM?

Three ways to condition a pretrained vision transformer on an external vector, compared against the constraint that matters most here: the pretrained weights must stay usable from step zero.

(a) Channel concatenation

Append K as extra input-image channels.

Corrupts the pretrained first layer — its weights were never fit to this channel count.

(b) FiLM-style modulation

Learn γ,β from K to rescale/shift feature maps directly (Perez et al. 2018).

Requires direct feature-map manipulation throughout the backbone — harder to gate to exactly zero everywhere at once.

(c) Token + attention

used

Encode K as one extra token in the sequence; let self-attention decide how much it matters.

Pretrained weights untouched — with the gate at zero, training can start from a fully **frozen** encoder.

Every result is a matched pair, not a single number

*Two models are fine-tuned from the **same** released Depth Anything V2 checkpoint, on the **same** data, seed, and schedule — differing only in whether the camera branch is present.*

proposed

+ camera intrinsics

`--model-variant revised --use-camera-intrinsics`

control

camera-blind

`--model-variant base` — identical otherwise

- Comparing **proposed vs. released DA2** answers "is the overall system better?"
- Comparing **proposed vs. matched control** — which differ *only* in access to K — isolates whether K specifically is the reason.
- Both comparisons are reported for **every** experiment in this talk.

released weights	40 epochs	lr 5×10^{-6}	frozen DINOv2	head ×10	grad clip 1.0

Two training regimes, two loss families

relative / "basic" models

Scale-shift-invariant loss, on disparity

Predictions carry no inherent scale, so both training and evaluation first solve a per-image least-squares alignment $D^{*}=s\cdot D+t$, then combine a pixel term with a multi-scale gradient-matching term.

$$L_{SSI} = (1-\alpha)\ L_{pixel} + \alpha\ L_{grad} \qquad \alpha \approx 0.5$$

$$T_{disp} = 1/\max(\ D^{GT}, d_{min}\ )$$

metric models

Scale-Invariant Log loss (SiLog)

Predictions are absolute depth in metres, unaligned at evaluation time — this is the regime where a global-scale signal like K has somewhere to matter, as §III shows.

$$L_{\text{SiLog}} = \sqrt{(1/N)\textstyle\sum_i d_i^2 - (\lambda/N^2)\left(\sum_i d_i\right)^2}$$

$d_i = \log y_i - \log \hat{y}_i$; λ controls how much global-scale error is forgiven — the campaign varies it $0.5 \rightarrow 0.3$.

III

The Experimental Campaign

Two consequential failures · a protocol pivot · from correlation to causation

This is where the thesis's evidence actually comes from. Early results exposed a contaminated test split and a collapsed ViT-L model. Fixing those failures, reporting two dead ends, and designing a targeted causal test made the final evidence trustworthy.

A skeptical, adversarial self-audit — not one training run

20+

independent training runs, 2 encoder sizes × 2 protocols

2

consequential failures diagnosed and fixed

2

honest negative results reported, not hidden

1

causal proof chain: correlation → mechanism → causation

The common thread: **a plausible training run is not evidence of a valid experiment.** A split audit exposed test leakage; output-value checks exposed a collapsed model despite a normal-looking loss curve.

Step 0: can this pipeline reproduce the paper?

Every comparison in this campaign depends on the evaluation pipeline being correct. Before trusting any of it, it was checked against Depth Anything V2's own published Table 2, zero-shot, with the released checkpoints.

BENCHMARK	OURS	PAPER
NYU · ViT-L (AbsRel / δ1)	0.043 / 0.979	0.045 / 0.979
NYU · ViT-S (AbsRel / δ1)	0.051 / 0.973	0.053 / 0.973
KITTI · ViT-L, no crop (AbsRel / δ1)	0.076 / 0.948	0.074 / 0.946
KITTI · ViT-S, no crop (AbsRel / δ1)	0.083 / 0.934	0.078 / 0.936

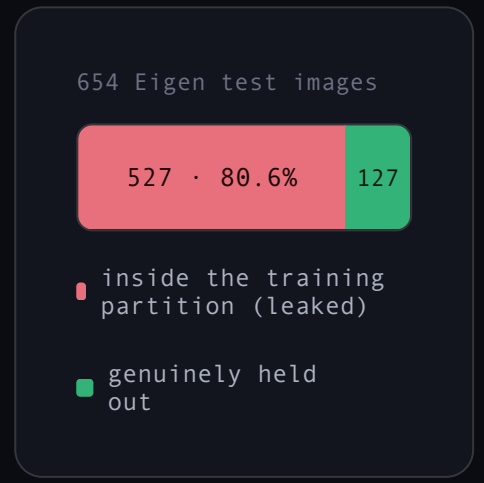
All four within normal reproduction tolerance. This check is a prerequisite for every table that follows, not a side note.

1 NYU Eigen test-set leakage

The training split random-shuffled all 1449 NYU images 80/20 **without first holding out** the standard 654-image Eigen test split used across the depth-estimation literature — including the DA2 paper itself.

Computed with the training seed: **527 of 654 test images were inside the training partition.** Every prior "NYU win" from an NYU-trained checkpoint in this project was contaminated.

Fixed — Eigen-654 excluded before the 80/20 split; 636 train / 159 val / 654 test, verified pairwise-disjoint



10 — CAMPAIGN — HONEST DEAD END

Multi-dataset zero-shot fine-tuning does not work

After fixing the leak, the leak-free zero-shot numbers were still far worse than the released baseline — for both arms equally.

RUN (ZERO-SHOT)	ABSREL	Δ1	VS. BASELINE
KITTI baseline (released DA2)	0.0829	0.9341	—
KITTI, camera-aware fine-tune	0.2139	0.6189	far worse
KITTI, matched control	0.2132	0.6179	far worse
NYU baseline (released DA2)	0.0508	0.9733	—
NYU, revised ViT-S (no NYU in training)	0.0657	0.9578	worse

Diagnosis

Fine-tuning a foundation model trained on ~62M images, on a few thousand small/synthetic images, destroys zero-shot robustness **regardless of the camera-intrinsics input** — 0.2139 vs. 0.2132 is noise. This strategy was abandoned; the campaign pivoted to a protocol where "better" is actually provable.

Switch to the protocol under which "better" is provable

Fine-tune on a benchmark's own training split; evaluate on its held-out test split; compare against the released model evaluated identically — the same benchmark-specific practice used by the released DA2 protocol.

```
train.py --encoder {vits|vitl} --datasets nyu --model-type {basic|metric}
  --min-depth 0.001 --max-depth {10|20} --epochs 40 --lr 5e-6
  --freeze-dinov2 --head-lr-multiplier 10 --grad-clip 1
  [proposed: --model-variant revised --use-camera-intrinsics]
  [control:  --model-variant base]
```

- **Start from the released checkpoint**, not from scratch — preserves the pretrained prior.
- **Frozen backbone, lr 5e-6** — only the head and camera encoder adapt, preventing the drift seen on the previous slide.

ViT-S: leak-free, and it wins

NYU Eigen-654, rawDepths GT, eigen crop, relative protocol · n = 654

MODEL	ABSREL↓	Δ1↑	RMSE↓	SQ_REL↓	LOG10↓
Released DA2 (baseline)	0.0508	0.9733	0.2414	0.0245	0.0222
Fine-tune, no intrinsics (control)	0.0507	0.9741	0.2088	0.0186	0.0218
+ camera intrinsics (proposed)	0.0501	0.9745	0.2064	0.0182	0.0216

Beats released DA2 on all five displayed metrics. Paired significance (n=654): vs. baseline — RMSE $p=5\times10^{-18}$, AbsRel n.s. ($p=0.09$). vs. matched control — AbsRel $p=0.005$, RMSE $p=3.5\times10^{-5}$ — at this scale, the win over the control is attributable to K itself.

2 ViT-L silently collapsed to zero

Scaling the identical recipe to ViT-L froze validation AbsRel at 0.346 from epoch 1 — the model learned to output **exactly zero everywhere**. The loss curve looked completely normal throughout.

Root cause

The pretrained model outputs **disparity** (inverse depth, values in the hundreds); training fed it **direct depth targets** (0–10 m). The loss's alignment scale is clamped positive (`min=0.001`), so it cannot fit an inverted relationship — and its gradient-matching term crushes the output toward zero, which is unrecoverable once the head's terminal ReLU is dead.

$$T_{disp} = 1/\max(D^{GT}, d_{min})$$

the fix — train on inverse-depth targets, matching the pretrained model's own output convention

0.001→1.0

dead-pixel fraction, within 18 steps

20–50×

lower loss after fix, verified before committing further GPU time

ViT-L beats baseline decisively — but ties its control

MODEL	ABSREL↓	Δ1↑	RMSE↓	SQ_REL↓
Released DA2 ViT-L (baseline)	0.0425	0.9789	0.2119	0.0195
Fine-tune, no intrinsics (control)	0.0378	0.9862	0.1922	0.0142
+ camera intrinsics (proposed)	0.0377	0.9861	0.1913	0.0143

vs. baseline: AbsRel $p=8.9\times10^{-6}$, RMSE $p=1.1\times10^{-9}$ — unambiguous. vs. matched control: statistical tie ($p=0.18$ / 0.50).

Why the tie? A capacity × data hypothesis

NYU is captured with a **single Kinect sensor** — K is constant across the entire training set, so the camera encoder has no variation to learn a mapping from. Worse: the existing pipeline randomly crops images **without** updating K to match — the encoder was training on values that were constant *and wrong*. This motivates the next fix.

Give K real variation — the fix helps everyone, but not via K

A zoom-crop augmentation resamples a random window and updates K **exactly**:

$$c_x' = (c_x - x_o) W_{net}/w_{win},\ c_y' = (c_y - y_o) H_{net}/h_{win}$$
$$f_x' = f_x \cdot W_{net}/w_{win},\ f_y' = f_y \cdot H_{net}/h_{win}$$

MODEL	ABSREL↓	Δ1↑	RMSE↓
Released DA2 ViT-L	0.0425	0.9789	0.2119
+ augmentation, no intrinsics	0.0359	0.9872	0.1849
+ augmentation + intrinsics	0.0362	0.9869	0.1881

Augmentation itself: a clear win for both arms ($p=1.2\times10^{-8}$ / 1.2×10^{-13}) — the best NYU numbers under the relative/basic protocol so far. But intrinsics is now marginally behind its control (AbsRel $p=0.0081$).

Why — a structural blind spot in the evaluation, not the model

Relative/"basic" evaluation aligns scale-and-shift **per image before scoring** — which removes exactly the **global metric scale** that a focal length disambiguates. At ViT-L capacity the visual encoder already extracts strong scale cues from content alone; this protocol removes global scale; the observed K benefit is small in this setting.

636 training images, fresh geometry every epoch

NYUv2 labelled subset · online augmentation · source training set: 636 → 636 images

01 — LEAK-FREE SPLIT

1,449
labelled RGB-depth pairs

654
Eigen test holdout

636
train · 159 validation

45.1% test holdout · remaining 795 split 80/20 · no test image enters training

02 — ONLINE GEOMETRY PER DRAW

20% full frame
80% random zoom-crop
s ∈ [1.0, 2.2] · fixed 4:3 window

636 views / epoch
25,440 over 40 epochs
sample draws across repeated images

03 — IMAGE + DEPTH + K STAY ALIGNED

Crop and resize RGB, depth and K together

$$s_x = 686 / w_{win}, \quad s_y = 518 / h_{win}$$
$$f_x' = s_x f_x, \quad f_y' = s_y f_y$$
$$c_x' = s_x (c_x - x_o), \quad c_y' = s_y (c_y - y_o)$$

640 × 480 → 686 × 518 network frame
Crop changes FoV; depth values remain unchanged

50% horizontal flip · RGB, depth and mask together; mirror c_x

Later focal jitter · $\alpha \in [0.625, 1.6]$
Scale f_x, f_y and depth by α; RGB pixels stay fixed

Matched arms share augmentation draws · validation/test stay fixed · Source: NYUTrainingDataset; train.py

Switch to metric depth — no alignment to hide behind

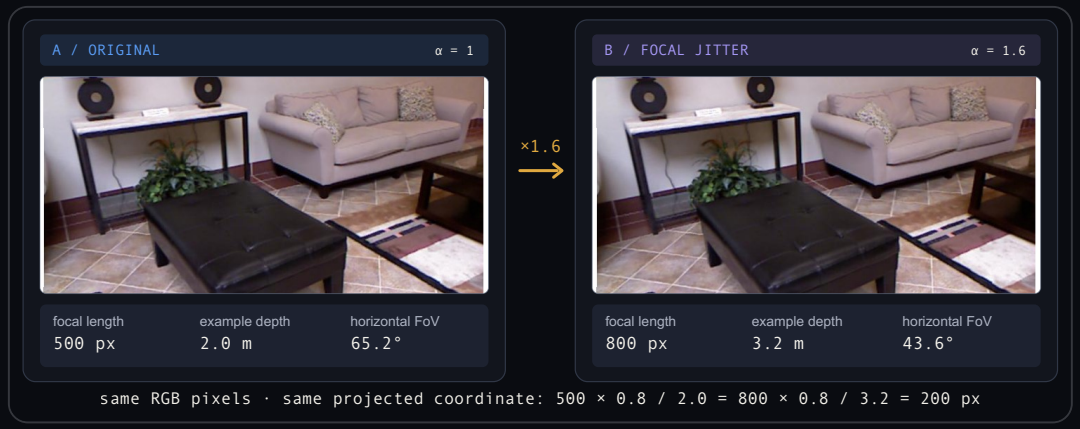
Absolute depth, no per-image scale-shift alignment · NYU Eigen-654

MODEL	ABSREL↓	Δ1↑	RMSE↓
Released DA2 metric (Hypersim)	0.2073	0.6919	0.6063
Round 1 (FoV ≈40–63°) · control	0.0796	0.9583	0.2919
Round 1 · + intrinsics	0.0791	0.9593	0.2882
Round 2 (FoV ≈31–63°, wider) · control	0.0826	0.9546	0.2998
Round 2 · + intrinsics	0.0817	0.9569	0.2944

Round 1 RMSE $p=4.9\times10^{-3}$, AbsRel n.s. → Round 2 RMSE $p=6.5\times10^{-5}$, AbsRel raw $p=0.040$; Holm $p=0.1595$ (n.s.). The effect grows as field-of-view variation widens — a dose-response pattern, and both fine-tunes beat the released metric model by 2.6× (AbsRel paired-t $p=4.8\times10^{-133}$).

Make scale mathematically unrecoverable from pixels alone

From the pinhole equation itself: scaling focal length by α **and** every scene depth by α produces the **identical image** — $u=fX/Z$ is invariant under $(f,Z)\to(\alpha f,\alpha Z)$.



Same pixels, different metric interpretation. α scales focal length and every depth label together; the displayed RGB image stays unchanged. The viewing ray is represented by the same projected pixel, while the camera's field of view changes with f.

$$\alpha \sim \exp(U(-\ln m, +\ln m)) \qquad m=1.6$$

What this does to the K-blind control

It now faces **literally identical pixels** paired with **contradictory** depth labels across training samples. Its best prediction is a loss-dependent compromise over the unresolved α distribution — an **irreducible error floor**, by construction.

What this does to the K-aware model

It can resolve α **exactly** by reading the (now-jittered) focal length off its own input. Using K stops being merely helpful and becomes **mathematically necessary** to avoid the control's failure mode.

- seed-locked pairs — bit-exact iteration-0 loss (1.725 = 1.725), so any difference is attributable only to K

All five metrics improve after Holm correction

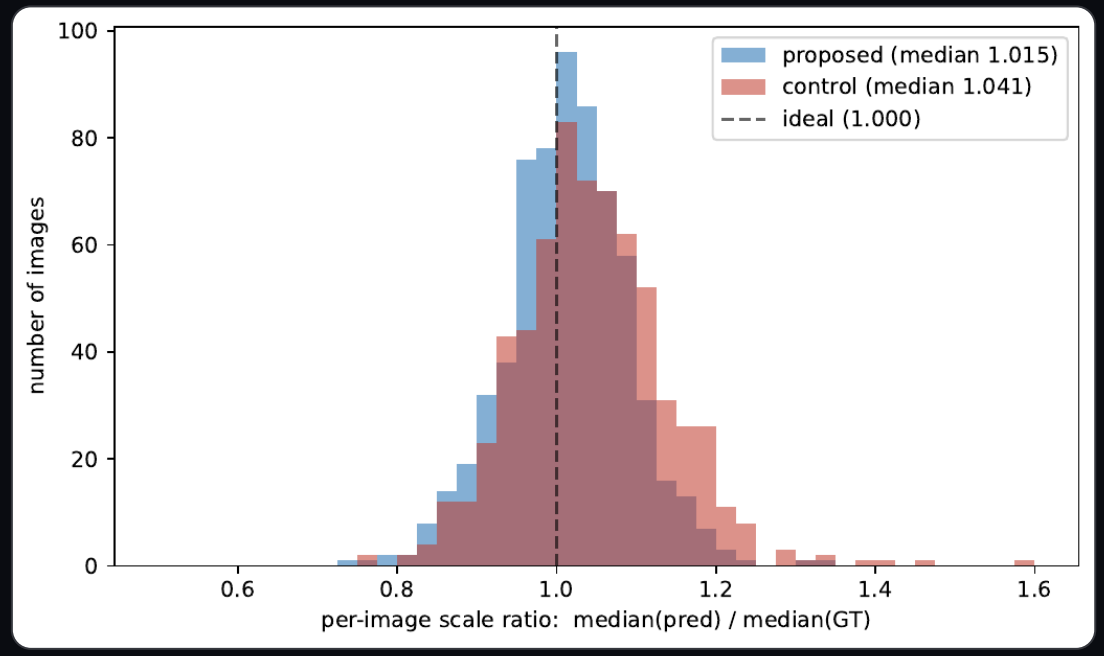
Standard NYU Eigen-654 metric evaluation, absolute depth, no alignment · n = 654

METRIC	+ INTRINSICS	CONTROL	WILCOXON P
AbsRel ↓	0.0849	0.1030	4.0×10^{-31}
RMSE ↓	0.3071	0.3431	1.0×10^{-24}
δ1 ↑	0.9520	0.9226	2.5×10^{-24}
SILog ↓	0.0942	0.1045	1.5×10^{-34}
log10 ↓	0.0363	0.0427	2.8×10^{-27}

A small price on absolute accuracy vs. the easier Round-1 setup (0.0849 vs. 0.0791) buys a jump in intrinsics evidence to a strong within-checkpoint difference that survives Holm correction.

The camera-aware model has a tighter scale-ratio distribution

Per-image ratio of predicted to ground-truth depth (median), focal-jitter round, all 654 eigen-test images



Proposed centers tightly at median ratio 1.015, over all 654 test images.

Reading the two distributions

The control distribution is both biased away from 1.0 and visibly wider (median **1.041**, sd 0.094 vs 0.078) — this is consistent with residual scale bias after ambiguous training.

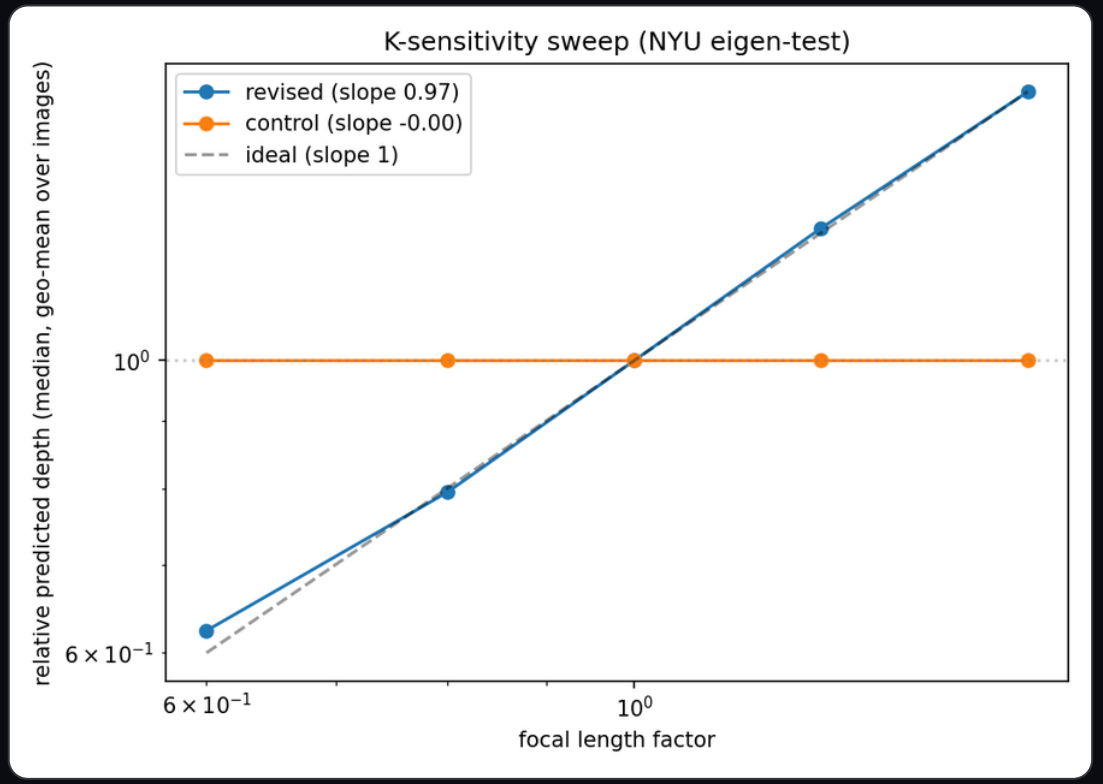
The proposed model's tight, centered distribution is descriptive evidence consistent with reading K lets it resolve α on a per-image basis, not just on average.

Is it the jitter, or the λ change? — It's the jitter

Round 3 changed both focal-jitter and `--silog-lambda` *(0.5→0.3) at once. A fourth seeded pair isolates jitter alone, λ held at 0.5.*

METRIC	+ INTRINSICS	CONTROL	WILCOXON P
AbsRel ↓	0.0834	0.1065	3.4×10^{-45}
RMSE ↓	0.2987	0.3497	7.5×10^{-40}
δ1 ↑	0.9554	0.9173	1.2×10^{-39}
SILog ↓	0.0918	0.1056	2.2×10^{-50}

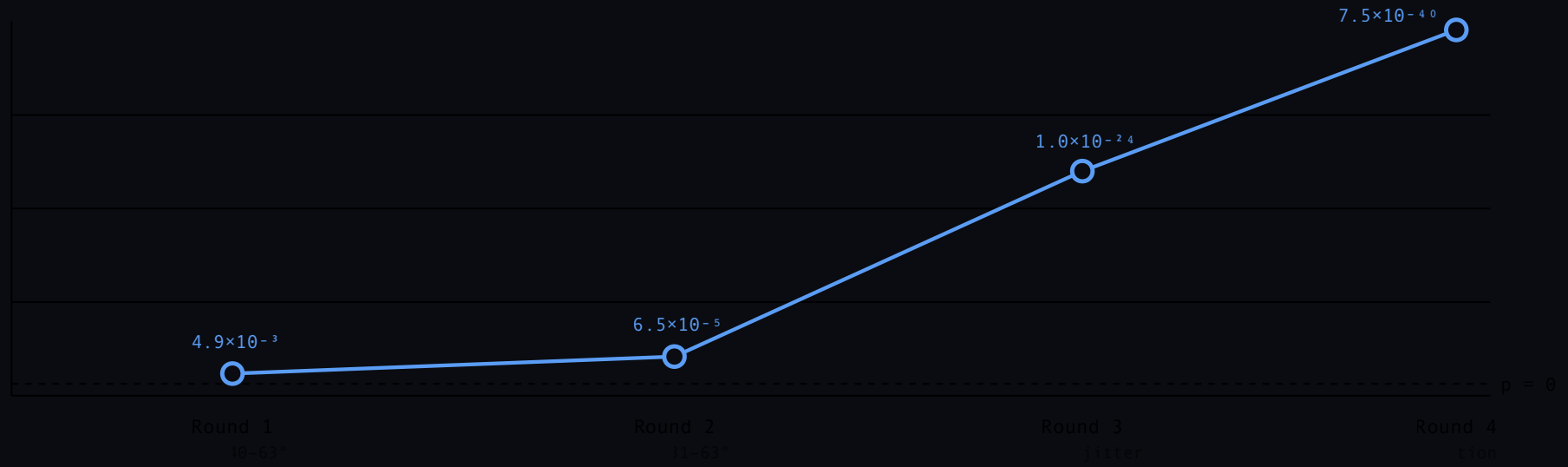
Slightly better than with λ=0.3 (AbsRel 0.0834 vs. 0.0849) — the λ change was unnecessary. Attribution is clean: the augmentation is the causal driver.



K-sweep slope for this pair: 0.97±0.03 — even closer to the ideal 1.0. Supporting λ=0.5 ablation; one pair, historical report.

The matched RMSE gap grows across these training settings

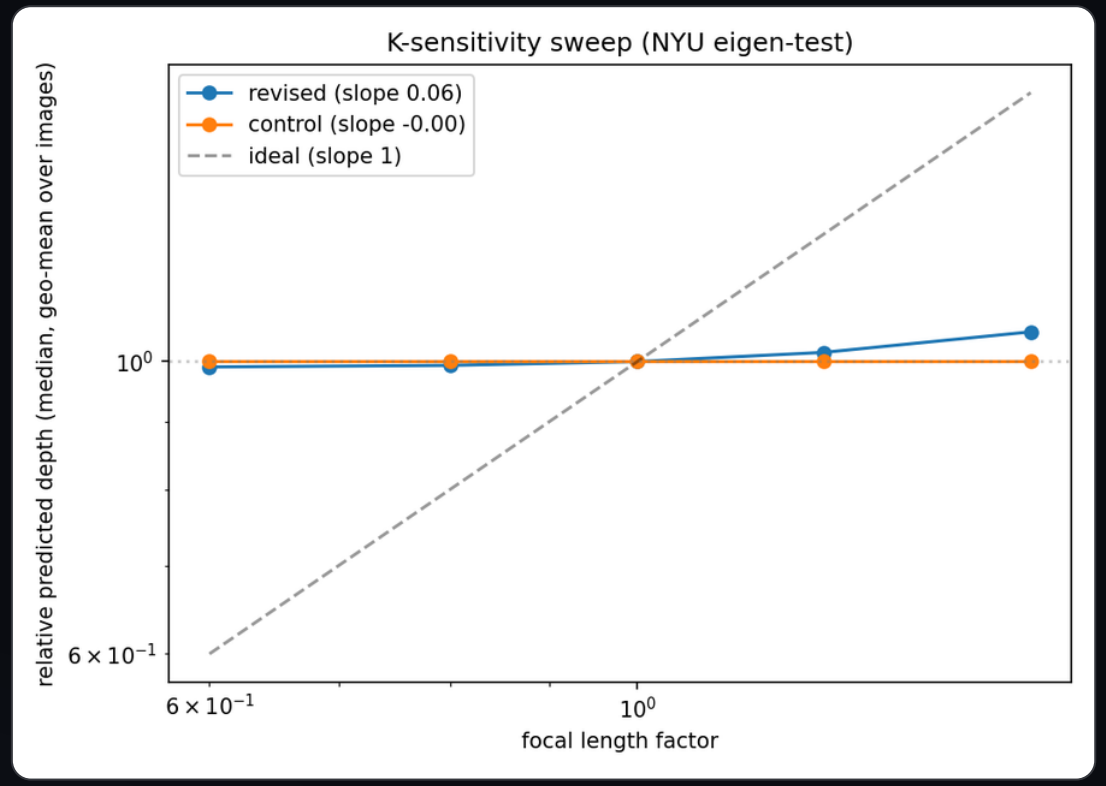
RMSE significance ($-\log_{10} p$) across four independently-trained matched training settings



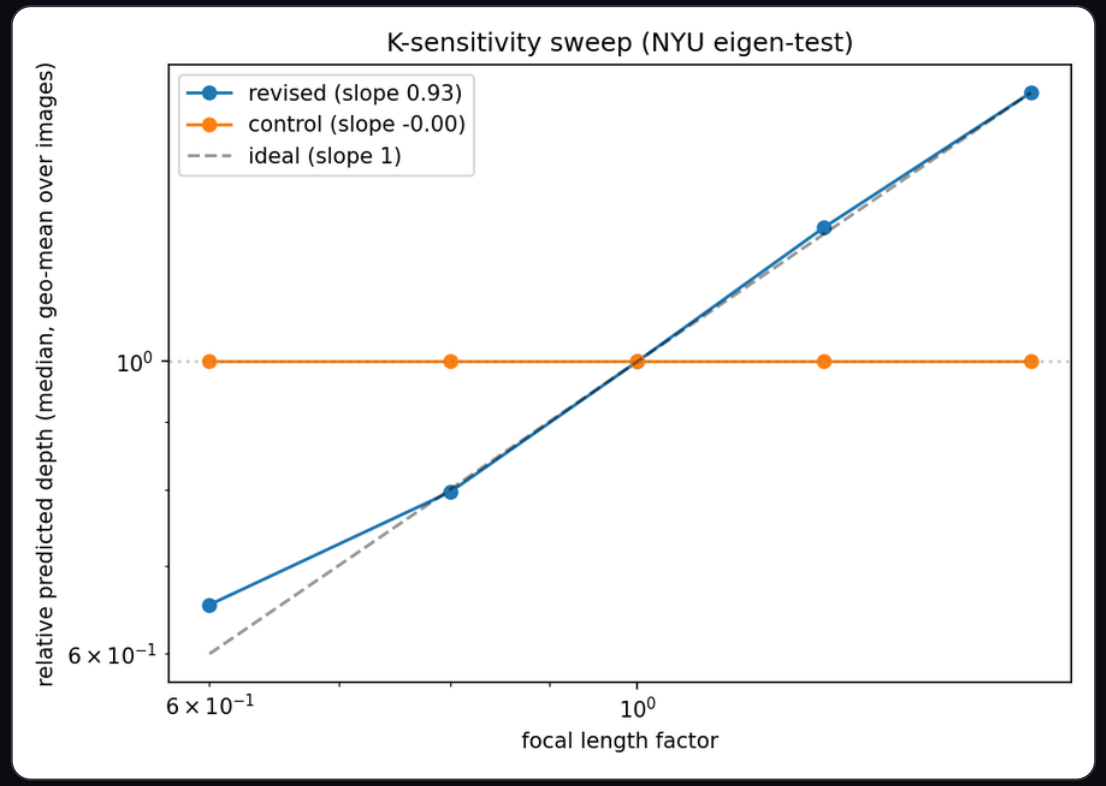
RMSE p-values: paired t-test in rounds 1–2; Wilcoxon in rounds 3–4. Different recipes are not independent-seed replications.

The K-sweep: rescale *f* at inference, watch the output move

20 test images; f_x, f_y scaled by {0.6, 0.8, 1.0, 1.25, 1.6}; log–log slope of predicted depth vs. injected scale. Ideal slope = 1.0 (exact pinhole rescaling); slope = 0 means K is ignored entirely.



Before focal-jitter. Proposed slope 0.06±0.01 · control 0.00. The architecture can read K but has no training pressure to.



After focal-jitter. Proposed slope 0.93±0.03 · control 0.00, unable to respond by construction.

Withhold or corrupt K — accuracy degrades on cue

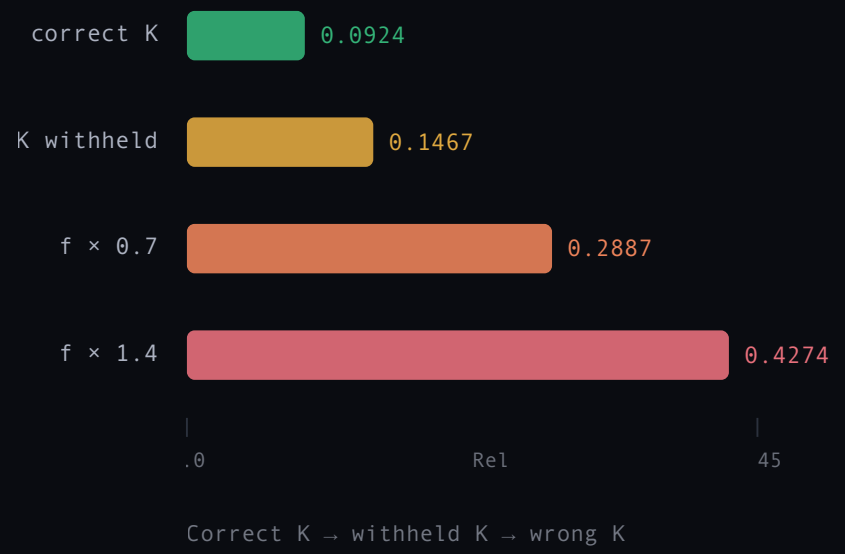
The round-3 proposed checkpoint, evaluated under four input conditions, full test set (n=654).

1. Withholding K is not decorative

AbsRel degrades **59%** (0.0924→0.1467) when K is withheld — a substantial share of the model's scale computation flows through this pathway.

2. The scale response is consistent with geometry

With measured slope s=0.93, the pinhole prediction for a wrong focal scale X is $|X^s-1|$: predicted ≈0.28 (X=0.7) and ≈0.37 (X=1.4) — observed 0.289 and 0.427. **This is approximate consistency, not an exact prediction of total AbsRel.**



Correct K generalizes across held-out field of view

A deterministic center zoom-crop changes the camera's field of view at inference. The proposed model keeps the advantage at every tested zoom, including the most severe crop.

Protocol boundary

All 654 Eigen images; metric AbsRel; no Eigen crop. These rows are an internal generalization diagnostic, so compare proposed versus control within each zoom.

CENTER ZOOM	+ INTRINSICS	CONTROL	GAP
1.0×	0.0924	0.1102	−0.0178
1.4×	0.0974	0.1100	−0.0127
1.8×	0.1065	0.1148	−0.0083

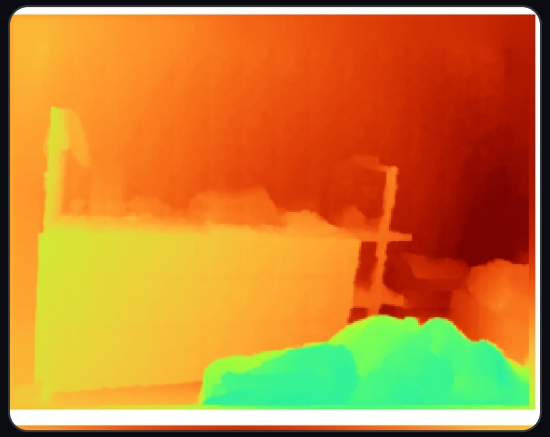
Lower is better. The proposed model wins at all three zoom levels.

A preregistered qualitative check

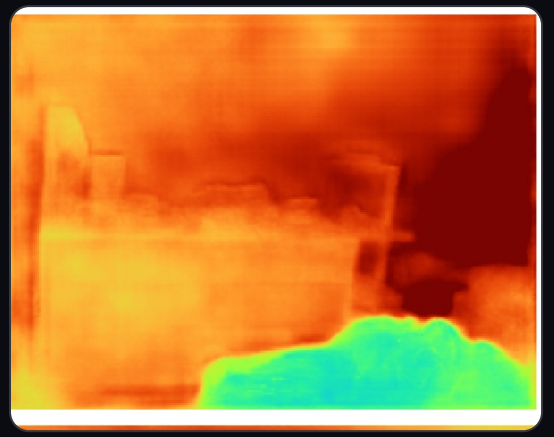
One of six locked NYU Eigen test-set examples, metric protocol



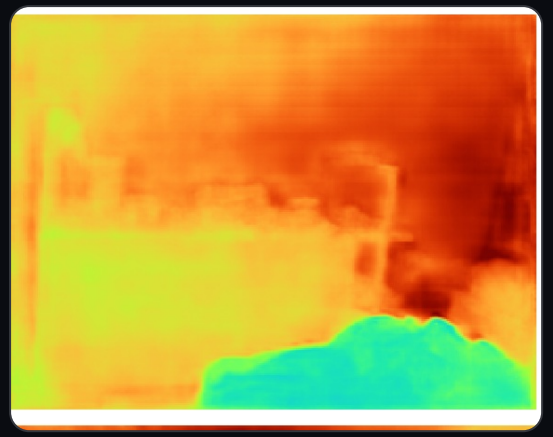
input image



ground truth



control (no K)



proposed (+K)

Illustrative visual check, not a leaderboard

The six-image set was selected before inference with seed **20260821** (selection hash prefix **8086cb124e40**) and rendered with common color scales. This displayed image is one of the six; the primary evidence remains the paired statistics, K-sweep, and K-knockout tests already shown.

CityScapes: what real, imperfect ground truth teaches

Real urban scenes, diverse lighting and weather, objects from near vehicles to distant buildings. Unlike NYU's active-sensor depth, CityScapes ground truth comes from **stereo matching + 3D reconstruction** — accurate near the camera on salient objects, but noisy and discontinuous in distant, weak-texture regions.

Why raw numbers here can mislead

RMSE and SiLog are especially sensitive to this GT noise; urban depth spans metres to tens of metres, so small relative errors become large absolute ones. A model that is more scale-stable will not necessarily look uniformly better on every metric here.

The explicit methodological stance

Direct numeric comparison of CityScapes results against other datasets is **not treated as valid** in this thesis. It is used strictly as a case study of model behaviour under complex, real, urban conditions — not as another row in a leaderboard.

Two consequential failures, one common lesson

1

Test-set leakage

527 of 654 NYU test images entered training. The split audit invalidated the earlier results.

2

Target/output mismatch

Disparity vs. depth → dead-ReLU collapse. The loss curve looked normal throughout.

Neither failure raised an exception. Each was surfaced only by an **independent end-to-end check** — auditing the split and monitoring validation/output statistics. This is why pipeline verification (§III.2) came first, not last.

20+

independent training runs across the full campaign

2

approaches tried and honestly abandoned (multi-dataset zero-shot; lowering head LR as a fix for collapse)

Six findings, condensed

#	FINDING	EVIDENCE
1	Proposed ViT-L beats released DA2 on both validated NYU protocols	Basic: AbsRel 0.0425→0.0377, paired-t $p=2.5\times10^{-8}$. Metric: 0.2073→0.0791, paired-t $p=4.8\times10^{-133}$
2	Net contribution of K is protocol-dependent	n.s. under relative/basic at ViT-L (alignment removes scale info); significant under metric
3	K's contribution shows a dose–response relationship	RMSE p: 4.9×10^{-3} → 6.5×10^{-5} → $\leq10^{-24}$, replicated across 4 pairs
4	The model demonstrably *reads* K, not just correlates	K-sweep slope 0.06 → 0.93–0.97 (ideal 1.0)
5	Reliance on K is causal, not correlational	Knockout: withholding K degrades AbsRel 59%; wrong K degrades accuracy; scale response is consistent with geometry
6	Observed K benefit differs between ViT-S and ViT-L	Small but significant at ViT-S even under basic protocol; not at ViT-L, capacity explanation is a hypothesis

Most of the evidence reproduces exactly; one table stays provisional

validated

Safe to present

Metric rounds 1–3: means and paired tests reproduce on NYU Eigen-654. Round-2 AbsRel loses significance after Holm correction; RMSE remains significant. Focal-jitter: **0.0849 vs 0.1030** AbsRel, all five metrics significant after Holm. K-sweep: **0.93±0.03**. K-knockout: AbsRel **0.0924→0.1467** without K and **0.2887 / 0.4274** under wrong focal length. Qualitative ratios: **1.015 vs 1.041**, SD **0.078 vs 0.094**, n=654. Table 4.9 and the FoV diagnostic also pass.

provisional

Do not headline yet

The cross-dataset §4.4 SiLog rows (74.38%, 54.78%, 63.90%, 70.42%) are **not confirmed**: fresh evaluation reproduces the baselines but not all four proposed means or gains. Keep them as provisional historical comparison rows until checkpoint provenance and units are reconciled.

Audit trail: §4.5.6 PASS · §4.5.7 PASS · Table 4.9 PASS · qualitative report present · §4.4.2 fresh validation reports mismatches in all four proposed rows.

IV

Discussion & Conclusion

Answering the research questions · contributions · limitations · future work

Back to the three questions

RQ1 Are common models stable across dataset and camera change?

No. The baseline is measurably sensitive on metric metrics across datasets and camera parameters (camera audit, cross-dataset SiLog differences) — instability that is real but often invisible to normalized evaluation.

RQ2 Can non-metric metrics conceal that instability?

Yes. AbsRel is GT-normalized and often hides scale-related change; RMSE consistently reveals it. Seen directly in this campaign (RMSE decisive where AbsRel is n.s.) and in why CityScapes numbers resist naive comparison.

RQ3 Does explicit K improve metric depth and reduce camera-sensitivity?

Yes — decisively under metric evaluation, protocol- and capacity-dependent under the relative/basic protocol. Proven not just statistically but **mechanistically** (K-sweep) and **causally** (K-knockout) — satisfying the causal-validity requirement raised alongside the original RQs.

Contributions — theoretical & practical

theoretical

1. Reframing scale ambiguity

As an inherent geometric limitation of image→3D mapping; in the limited-data regime, explicit camera geometry provides a direct signal for resolving camera-scale ambiguity.

2. Metric vs. non-metric evaluation

A conceptual framework showing normalized-metric evaluation alone can give an incomplete, sometimes misleading, picture of true model performance.

practical

3. A minimal, compatible solution

Preserves the base architecture; adds K as one gated token — no large architectural change, no extra data collection required.

4. Works with a fully frozen encoder

Meaningful metric improvement achieved without touching the pretrained feature-extraction weights at all — a low-risk adoption path for practitioners.

Contributions — methodological & diagnostic

Each demonstrated at length in §III — this is the formal inventory.

5. Found & fixed the **NYUv2 split leak** (>80% test contamination) — a precondition for any valid comparison.

6. Diagnosed the **disparity/depth target-space collapse** — silent, invisible in the loss curve.

7. **Camera-aware geometric augmentation** — exact-K zoom-crop, from a single-camera dataset.

8. Showed the **relative protocol is structurally blind** to K's main contribution (absolute scale).

9. **Focal-jitter augmentation** — makes scale recovery from pixels alone provably impossible without K.

10. The **K-sweep diagnostic** — a general-purpose mechanistic test, more informative than aggregate metrics or the gate value.

11. **Seed-locked paired design**, verified bit-exact — attributes any difference to K alone.

12. Documented **two consequential failure patterns** in deep-learning pipelines as a general lesson.

Limitations, stated precisely

1. **Dependence on K's accuracy.** K-knockout quantified this: full removal ≈59% AbsRel increase; 30–40% focal distortion causes several-times baseline error. Realistic (small, non-adversarial) calibration noise was not tested.
2. **Ground-truth quality varies across datasets** — generation method, depth range, and uniformity all differ, affecting absolute error interpretation (see CityScapes, §III).
3. **Single base architecture.** Implemented and evaluated only on Depth Anything V2; conceptually architecture-independent, but generalization to other encoders/decoders is untested.
4. **Static, single-image scope.** Temporal and multi-view geometric constraints are unused — a deliberate choice to isolate K's role, but it limits applicability.
5. **The controlled study is NYU-only**, with camera diversity *simulated* via augmentation rather than genuine multi-camera capture; round-2's AbsRel significance was borderline (raw p=0.040; Holm p=0.1595, n.s.) even where RMSE remained significant; generalizing to outdoor, large-depth-range benchmarks (e.g. KITTI) is unstarted.
6. **Single seed per arm.** Seed-locking gives strong within-pair causal inference, but training-process variance across independent seeds was not separately estimated.

Where this goes next

1. **Robustness to imprecise K** — study behaviour under realistic calibration noise; design uncertainty-aware handling.
2. **Joint depth & intrinsics estimation** — estimate or refine K jointly with depth, instead of assuming it's known.
3. **Genuinely multi-camera training data** — the dose-response pattern predicts an even larger, more stable K contribution; also sweep the token-injection layer and depth.
4. **Multi-seed statistical robustness** — repeat each paired arm across independent seeds, reporting mean/std alongside the paired image-level tests.
5. **Multi-image / video** — combine temporal and multi-view constraints with explicit K conditioning.
6. **Architecture generalization** — test whether the gains transfer to other encoder/decoder families, or are specific to this one.

THANK YOU

Questions & Discussion

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